Problem Set **#1**

P1. Answer with reason. If the intrinsic carrier concentration (n_i) in Si, Ge and GaAs samples at room temperature are compared, which of the following statements is true.

(a) n_i of Si > n_i of Ge > n_i of GaAs
(b) n_i of GaAs > n_i of Si > n_i of Ge
(c) n_i of Ge > n_i of Si > n_i of GaAs

P2. Island-X is very special. Here only twins are born, one male and the other female. Also, on this island, partners (a male and a female) die simultaneously. On this island, 1,210 pairs of twins are born every year. The number of couples dying per year is 0.001% of the product of the male population (m) and female population (f). (The logic is as follows: The number of deaths is proportional to the male population. It is also proportional to the female population. Hence, the number of deaths is proportional to the product of the male and female populations.)

(a) If the population of the island was stable over many years, what are the values of m and f?

(b) The people of the island decided to introduce birth control measures and brought down the birth rate to 1,000 pairs of twins/year. After some years, the population of the island stabilized. What are the new values of m and f?

(c) Relate the above to generation, recombination and intrinsic carrier concentration in a semiconductor.

(d) Then suddenly in one year, 1,000 males invaded the island. They slowly integrated with the other inhabitants of the island. When the population stabilized after some years, what are the new values of m and f? Note that the birth rate is still 1,000 pairs of twins/year.

(e) If 10,000 males had invaded the island, what would have been the new values of m and f after the population stabilized?

(f) What is the product of m and f in (b), (d) and (e)? What can you conclude?

(g) If, after an invasion, the value of m had stabilized at 12,500, what would be f and the number of invaders?

(h) Relate (d) – (f) to the effect of doping in semiconductors.

P1. Answer with reason. If the intrinsic carrier concentration (n_i) in Si, Ge and GaAs samples at room temperature are compared, which of the following statements is true.

(a) n_i of Si > n_i of Ge > n_i of GaAs
(b) n_i of GaAs > n_i of Si > n_i of Ge
(c) n_i of Ge > n_i of Si > n_i of GaAs

E_g GaAs = 1.42 eV E_g Si = 1.12eV E_g Ge = 0.66eV

$\Rightarrow n_i = \sqrt{N_c N_v}\, e^{-E_g/2kT} \Rightarrow (n_i)_{Ge} > (n_i)_{Si} > (n_i)_{GaAs}$

P2. Island-X is very special. Here only twins are born, one male and the other female. Also, on this island, partners (a male and a female) die simultaneously. On this island, 1,210 pairs of twins are born every year. The number of couples dying per year is 0.001% of the product of the male population (m) and female population (f). (The logic is as follows: The number of deaths is proportional to the male population. It is also proportional to the female population. Hence, the number of deaths is proportional to the product of the male and female populations.)
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(b) The people of the island decided to introduce birth control measures and brought down the birth rate to 1,000 pairs of twins/year. After some years, the population of the island stabilized. What are the new values of m and f?
(c) Relate the above to generation, recombination and intrinsic carrier concentration in a semiconductor.
(d) Then suddenly in one year, 1,000 males invaded the island. They slowly integrated with the other inhabitants of the island. When the population stabilized after some years, what are the new values of m and f? Note that the birth rate is still 1,000 pairs of twins/year.
(e) If 10,000 males had invaded the island, what would have been the new values of m and f after the population stabilized?
(f) What is the product of m and f in (b), (d) and (e)? What can you conclude?
(g) If, after an invasion, the value of m had stabilized at 12,500, what would be f and the number of invaders?
(h) Relate (d) – (f) to the effect of doping in semiconductors.

x → intrinsic semiconductor

Birthrate = b

death p.y = $10^{-5}mf$

$b = 10^{-5}mf$

m = f

(a) $1210 = 10^{-5}mf \Rightarrow m = f = 11\times10^3$

(b) $1000 = 10^{-5}mf \Rightarrow m = f = 10^4$

(c) $m \leftrightarrow n$ $f \leftrightarrow p$ $mf = \text{const} \leftrightarrow np = n_i^2$
birth of twins ↔ e^--h^+ generation
death of couple ↔ e^--h^+ recombination

(d) $10^{-5}(f+1000)f = 1000 \Rightarrow f(f+10^3) = 10^8 \Rightarrow f = 9513 \approx 9500$
($m = f+1000$) $m = 10513 \approx 10500$

(e) $10^{-5}(f+10^4)f = 10^3 \Rightarrow f(f+10^4) = 10^8 \Rightarrow f = 6200$
$m = 16200$

(f) 10^8 (mf only depends on birth rate, not doping)

(g) $10^{-5}\times12500\times f = 10^3 \Rightarrow f = 8000 \Rightarrow$ 2500 people invaded

(h) male invasion ↔ n-type doping
↑ recombination ↔ ↑ carrier recombination
$mf = \text{const} \leftrightarrow np = n_i^2$

EE3001: Solid State Devices
TAs: Alias George (ee24d402@smail.iitm.ac.in) and Devendar K (ee24s015@smail.iitm.ac.in)
Assignment #2

Try yourself to solve the problems with detailed steps. Take a limited time quiz to test your understanding on related concepts.

Problems will be discussed during the tutorial class. You need to contact TAs or teacher if you are unable to do anything useful by yourself or with the help of your friends.

If needed, assume the following values by default unless otherwise specified: Si is the default semiconductor; SiO_2 is the default oxide; T @room temperature is 300 K; n_i@room temp. = $10^{10} cm^{-3}$; k_BT=0.026 eV (at room temperature); $E_{G,Si}$ = 1.12 eV and temperature independent; at 300 K, N_c=$2.8 \times 10^{19} cm^{-3}$ and N_v=$1.02 \times 10^{19} cm^{-3}$ $K = 8.62 \times 10^{-5}$ eV

1. (a) If N_c represents the density of states in a strip αkT wide near the edge of the conduction band, determine the value of α.

   (b) Obtain $(E_c - E_F)$ at room temperature for electron concentrations of (i) 5×10^{14} cm^{-3}, (ii) 5×10^{17} cm^{-3} and (iii) 10^{19} cm^{-3} in Si. Show the positions of E_F in the energy bandgap. Comment on the applicability of Boltzmann approximation in various cases.

2. (a) For a non-degenerate Si sample at room temperature, E_c - E_f = $4.1 k_BT$. Calculate the probability of finding an electron at an energy level of k_BT above the conduction band.

   (b) Find the doping concentration in the Si sample at T=300 K if all the dopant atoms are ionized.

3. In a semi-log axis the plot of $1/k_BT$ (linear x-axis in per eV) versus $n_i/\sqrt{N_cN_v}$ (log_e y-axis) for a semiconductor sample appears to be linear with a slope of -0.71 near T=300 K. What is the band-gap of the semiconductor?

4. The average kinetic energy of electrons in the conduction band is given by the following relation

$$< K.E >= \frac{\int_{E_c}^{\infty} (E - E_c) D(E) f(E) dE}{\int_{E_c}^{\infty} D(E) f(E) dE}$$

   For a non-degenerate semiconductor in which $f(E)$ can be approximated by the Maxwell-Boltzmann distribution, if the average kinetic energy is given by $\lambda\, k_BT$, find the value of λ. Here D(E) denotes the density of states in conduction band.

5. (a) For a pure Si sample, if the temperature is increased from 300 K to 400 K, find the ratio of the intrinsic carrier concentration, $n_i(400K)/n_i(300K)$.

(a) If N_c represents the density of states in a strip αkT wide near the edge of the conduction band, determine the value of α.

$$\int_{E_c}^{E_c + \alpha KT} D_c(E)\, dE = N_c$$

$$\frac{2}{4\pi}\left(\frac{2m_e^*}{h^2}\right)^{3/2} \cdot \frac{2}{3} \cdot \alpha^{3/2} (KT)^{3/2} = 2\left(\frac{2\pi m_e^* KT}{h^2}\right)^{3/2} \Rightarrow \alpha = \left(9\pi/16\right)^{1/3} = 1.209$$

(b) Obtain $(E_c - E_F)$ at room temperature for electron concentrations of (i) 5×10^{14} cm^{-3}, (ii) 5×10^{17} cm^{-3} and (iii) 10^{19} cm^{-3} in Si. Show the positions of E_F in the energy bandgap. Comment on the applicability of Boltzmann approximation in various cases.

$KT = 26meV$

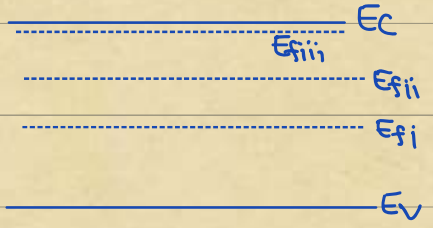
$N_C = 2.8 \times 10^{19}$

$n = N_C \exp\left(\frac{E_F - E_C}{KT}\right) \Rightarrow E_C - E_f = \ln\left(\frac{N_C}{n}\right) \cdot KT$

(i) $E_c - E_f = 0.28eV$ (valid)

(ii) $E_C - E_f = 0.104\,eV$ (valid)

(iii) $E_C - E_F = 0.027\,eV$ (not valid)



(a) For a non-degenerate Si sample at room temperature, E_c - E_f = $4.1k_BT$. Calculate the probability of finding an electron at an energy level of k_BT above the conduction band.

$E = E_C + kT \qquad E - E_f = 5.1KT \qquad f(E) = \frac{1}{1 + e^{(E-E_f)/KT}} = 6.06 \times 10^{-3}$

(b) Find the doping concentration in the Si sample at T=300 K if all the dopant atoms are ionized.

$E_C - E_f = KT \ln\left(\frac{N_C}{n}\right) \Rightarrow \frac{N_C}{n} = 60.34 \Rightarrow n = 4.64 \times 10^{17} = N_D$

3. In a semi-log axis the plot of $1/k_BT$ (linear x-axis in per eV) versus $n_i/\sqrt{N_cN_v}$ (log_e y-axis) for a semiconductor sample appears to be linear with a slope of -0.71 near T=300 K. What is the band-gap of the semiconductor?

$n_i = \sqrt{N_cN_v}\exp\left(-\frac{E_g}{2KT}\right) \Rightarrow \ln\left(\frac{n_i}{\sqrt{N_cN_v}}\right) = \left(-\frac{E_g}{2}\right)\frac{1}{k_BT}$

$\Rightarrow -0.71 = -\frac{E_g}{2} \Rightarrow E_g = 1.42eV$

4. The average kinetic energy of electrons in the conduction band is given by the following relation

$$< K.E >= \frac{\int_{E_c}^{\infty}(E - E_c)D(E)f(E)dE}{\int_{E_c}^{\infty} D(E)f(E)dE} = \frac{\int_0^{\infty}\mu\sqrt{\mu}e^{-\beta\mu}d\mu}{\int_0^{\infty}\sqrt{\mu}e^{-\beta\mu}d\mu} = \frac{\Gamma(5/2)(1/\beta)^{5/2}}{\Gamma(3/2)(1/\beta)^{3/2}} = \frac{3}{2\beta}$$

For a non-degenerate semiconductor in which $f(E)$ can be approximated by the Maxwell-Boltzmann distribution, if the average kinetic energy is given by $\lambda\, k_BT$, find the value of λ. Here D(E) denotes the density of states in conduction band.

$f(E) = e^{-(E-E_f)\beta} \qquad D(E) = A\sqrt{E - E_c} \qquad < K.E > = 3/2\, k_BT \Rightarrow \lambda = 1.5$

(a) For a pure Si sample, if the temperature is increased from 300 K to 400 K, find the ratio of the intrinsic carrier concentration, $n_i(400K)/n_i(300K)$.

$n_i \propto T^{3/2}e^{-E_g/2KT} \Rightarrow \frac{n_i(400)}{n_i(300)} = \left(\frac{4}{3}\right)^{1.5}\exp\left[\frac{-1.12}{2 \times 8.62 \times 10^{-5}}\left(\frac{1}{400} - \frac{1}{300}\right)\right] = 345.6$

(b) At 400 K, what are the values of n_i for (i) Ge (E_g = 0.66 eV, $N_c = 10^{19}\ cm^{-3}$, $N_v = 6 \times 10^{18}\ cm^{-3}$) and (ii) GaAs ($E_g$ = 1.42 eV, $N_c = 4.7 \times 10^{17}\ cm{-}3$, $N_v = 7 \times 10^{18}\ cm^{-3}$)? Note that given values of N_c and N_v are in 300 K.

6. (a) A Si sample already doped with acceptor atoms ($N_A = 10^{14}\ cm^{-3}$) is further doped with donor atoms ($N_D\ cm^{-3}$). Calculate the free carrier concentration (n and p) for (i) $N_D = N_A$; (ii) $N_D = 5N_A$; (iii) $N_A = 3N_D$;

   (b) Check and comment if the calculations of n and p are simplified when $|N_D - N_A| \ll 2n_i$ or $|N_D - N_A| \gg 2n_i$.

   (c) Plot the concentration of n and p as a function of N_D (varying from $10^{13}\ cm^{-3}$ to $10^{17}\ cm^{-3}$) if N_A is fixed at $10^{15}\ cm^{-3}$.

   (d) If the Si sample in (a) has no donor doping, calculate the ratio, p/n at (i) 300 K, (ii) 400 K, (iii) 500 K and (iv) 600 K and plot corresponding $E_F - E_i$ as a function of temperature.

(b) At 400 K, what are the values of n_i for (i) Ge (E_g = 0.66 eV, $N_c = 10^{19}\ cm^{-3}$, $N_v = 6 \times 10^{18}\ cm^{-3}$) and (ii) GaAs ($E_g$ = 1.42 eV, $N_c = 4.7 \times 10^{17}\ cm{-}3$, $N_v = 7 \times 10^{18}\ cm^{-3}$)? Note that given values of N_c and N_v are in 300 K.

(i) $n_i = \sqrt{10^{19} \times 6 \times 10^{18}}\ e^{-0.66/2 \times 400 \times 8.62 \times 10^{-5}} \times (4/3)^{1.5} = 8.32 \times 10^{14}/cm^3$

(ii) $n_i = \sqrt{4.7 \times 10^{17} \times 7 \times 10^{18}}\ e^{-1.42/2 \times 400 \times 8.62 \times 10^{-5}} (4/3)^{1.5} = 3.19 \times 10^{9}/cm^3$

(a) A Si sample already doped with acceptor atoms ($N_A = 10^{14}\ cm^{-3}$) is further doped with donor atoms ($N_D\ cm^{-3}$). Calculate the free carrier concentration (n and p) for (i) $N_D = N_A$; (ii) $N_D = 5N_A$; (iii) $N_A = 3N_D$;

(i) $N_A = N_D \Rightarrow n = p = n_i = 10^{10}/cm^3$

(ii) $N_D = 5 \times 10^{14} \Rightarrow p + 5 \times 10^{14} = n + 10^{14} \Rightarrow n - p = 4 \times 10^{14} \Rightarrow n \approx 4 \times 10^{14}/cm^3$

$p \approx n_i^2/n = 2.5 \times 10^{5}/cc$

(iii) $N_D = N_A/3 \Rightarrow p - n = 2/3 \times 10^{14} \Rightarrow p = 2/3 \times 10^{14} \Rightarrow n = 1.5 \times 10^{6}$

(b) Check and comment if the calculations of n and p are simplified when $|N_D - N_A| \ll 2n_i$ or $|N_D - N_A| \gg 2n_i$.

$|N_D - N_A| << 2n_i \Rightarrow n \approx p \approx n_i$ $\quad |N_D - N_A| >> 2n_i \rightarrow$ majority $\approx |N_D - N_A|$

$\rightarrow$ minority $\approx \dfrac{n_i^2}{\text{majority}}$

(d) If the Si sample in (a) has no donor doping, calculate the ratio, p/n at (i) 300 K, (ii) 400 K, (iii) 500 K and (iv) 600 K and plot corresponding $E_F - E_i$ as a function of temperature.

$N_D = 0$ $\quad N_A = 10^{14} \Rightarrow p = 10^{14}$

$n_i(T = T_1) = n_i(T = 300K) \times (T_1/300)^{1.5}$

$\times \exp\left(\dfrac{-1.12}{2 \times 8.62 \times 10^{-5}} \left(\dfrac{1}{T} - \dfrac{1}{300}\right)\right)$

$T = 300K \Rightarrow p/n = 10^{14}/10^{6} = 10^{8}$

$T = 400K \Rightarrow p/n = 839.232$

$T = 500K \Rightarrow p/n = 0.646$

$T = 600K \Rightarrow p/n = 0.005$

$E_f - E_i = -KT \ln(p/n_i)$

low $T \Rightarrow E_f \rightarrow E_v$

high $T \Rightarrow E_f \rightarrow E_i$

EE3001: Solid State Devices
TAs: Alias George (ee24d402@smail.iitm.ac.in) and Devendar K (ee24s015@smail.iitm.ac.in)
Assignment #3

Try yourself to solve the problems with detailed steps. Take a limited time quiz to test your understanding on related concepts.

Problems will be discussed during the tutorial class. You need to contact TAs or teacher if you are unable to do anything useful by yourself or with the help of your friends.

If needed, assume the following values by default unless otherwise specified: Si is the default semiconductor; SiO_2 is the default oxide; T @room temperature is 300 K; n_i@room temp. = $10^{10} cm^{-3}$; k_BT=0.026 eV (at room temperature); $E_{G,Si}$ = 1.12 eV and temperature independent; at 300 K, N_c=$2.8 \times 10^{19} cm^{-3}$ and N_v=$1.02 \times 10^{19} cm^{-3}$

1. (2 points) Consider two semiconductor materials (M1 and M2) with the following properties: electron affinity = 4.1 eV (M1), 4.2 eV (M2); Bandgap = 1.4 eV (M1); 1 eV (M2) and is p-type with $E_i - E_f$ = 0.2 eV (M1), n-type with $E_f - E_i$ = 0.4 eV (M2).

   (a) Draw the energy band diagram for this junction.

   (b) What is the built-in (contact) potential across a junction of these two materials? Which side will be at higher potential?

   (c) What is the direction of the electrostatic field? Will the electric field be constant or varying with position?

2. (2 points) A current of 2 mA flows through a bar of uniformly doped n type silicon with a cross-sectional area of $2mm \times 2mm$ and a length of 1 cm when it is connected to a 3 V battery at 300 K.

   (a) Calculate the electron and hole concentrations (in per cm^3) in the bar assuming electron mobility to be 1350 $cm^2V^{-1}s^{-1}$ and the hole mobility 450 $cm^2V^{-1}s^{-1}$.

   (b) Draw the energy band diagram for the Si bar under this biased condition.

3. (2 points) If at an acceptor doping concentration of N_A, a sample of Si attains maximum resistivity at 300 K, considering complete ionization of dopant atoms find the ratio N_A/n_i. (μ_n = 1350 $cm^2V^{-1}s^{-1}$, μ_p = 450 $cm^2V^{-1}s^{-1}$)

4. (2 points) A Boron doped (10^{16} atoms/cm^3) Si sample has resistivity of 0.1 Ω-cm. Calculate the doping concentration of Phosphorus atoms if it is additionally used to reduce the resistivity of this doped sample by 70%. Assume that the electron mobility is three times of the hole mobility throughout.

5. (2 points) A particular semiconductor sample is doped with 2×10^{16} donor atoms/cm^3. The intrinsic carrier concentration (n_i) in the semiconductor is $2 \times 10^{14}/cm^3$ at 300 K.

1. (2 points) Consider two semiconductor materials (M1 and M2) with the following properties: electron affinity = 4.1 eV (M1), 4.2 eV (M2); Bandgap = 1.4 eV (M1); 1 eV (M2) and is p-type with $E_i - E_f$ = 0.2 eV (M1), n-type with $E_f - E_i$ = 0.4 eV (M2).

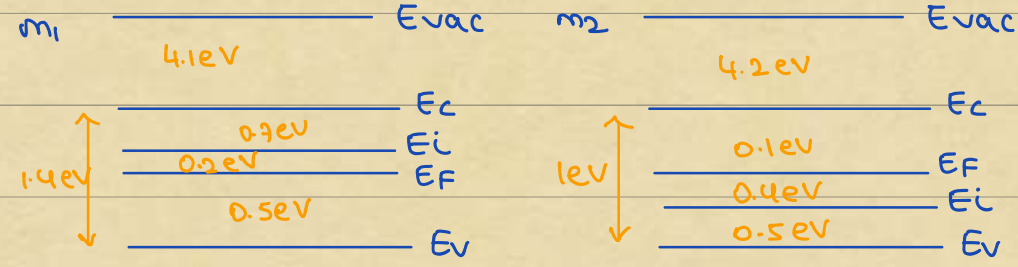
   (a) Draw the energy band diagram for this junction.

   (b) What is the built-in (contact) potential across a junction of these two materials? Which side will be at higher potential?

   (c) What is the direction of the electrostatic field? Will the electric field be constant or varying with position?

$E_i \approx \frac{E_C + E_V}{2}$

(a)



(b) 5.0 − 4.3 = 0.7 eV M_1 is @ higher potential due to higher work function ($E_{vac} - E_f$)

(c) E flows from M_1 to M_2

$$\frac{d^2\psi}{dx^2} = \frac{-\rho}{\epsilon} \Rightarrow \frac{dE}{dx} = \frac{-\rho(x)}{\epsilon} \rightarrow \text{varies with position}$$

2. (2 points) A current of 2 mA flows through a bar of uniformly doped n type silicon with a cross-sectional area of $2mm \times 2mm$ and a length of 1 cm when it is connected to a 3 V battery at 300 K.

   (a) Calculate the electron and hole concentrations (in per cm^3) in the bar assuming electron mobility to be 1350 $cm^2V^{-1}s^{-1}$ and the hole mobility 450 $cm^2V^{-1}s^{-1}$.

   (b) Draw the energy band diagram for the Si bar under this biased condition.

$A = 4 \times 10^{-2} cm^2$

$E = V/L = 3 V/cm$

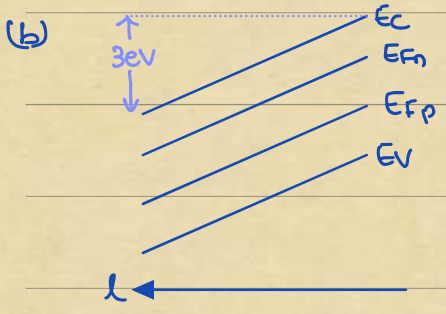
$J = I/A = 0.05 A/cm^2$

(a) $J = nq\mu_n E + pq\mu_p E$

$5 \times 10^{-2} = 1.6 \times 10^{-19} (1350n + 450p) \times 3$ $3n + p = 2.314 \times 10^{14}$

$n = 7.71 \times 10^{13}$; $p = 1.3 \times 10^{6}$

(b)



3. (2 points) If at an acceptor doping concentration of N_A, a sample of Si attains maximum resistivity at 300 K, considering complete ionization of dopant atoms find the ratio N_A/n_i. ($\mu_n = 1350\ cm^2V^{-1}s^{-1}$, $\mu_p = 450\ cm^2V^{-1}s^{-1}$)

$p = n + N_A$

$n(n+N_A) = n_i^2$

$\sigma = q(n\mu_n + p\mu_p) = q[n(\mu_n+\mu_p) + N_A\mu_p]$

$= q\left[n(\mu_n+\mu_p) + \left(\frac{n_i^2}{n} - n\right)\mu_p\right]$

$= q\left[n(\mu_n) + \frac{1}{n}(n_i^2\mu_p)\right]$

$\Rightarrow n\mu_n = \mu_p\frac{n_i^2}{n} \Rightarrow n = n_i\sqrt{\frac{\mu_p}{\mu_n}} = \frac{n_i}{\sqrt{3}}$

$N_A = \frac{n_i^2}{n} - n = \frac{n_i^2}{n_i/\sqrt{3}} - n_i/\sqrt{3} = \frac{2n_i}{\sqrt{3}} \Rightarrow \frac{N_A}{n_i} = \frac{2}{\sqrt{3}}$

4. (2 points) A Boron doped (10^{16} atoms/cm^3) Si sample has resistivity of 0.1 Ω-cm. Calculate the doping concentration of Phosphorus atoms if it is additionally used to reduce the resistivity of this doped sample by 70%. Assume that the electron mobility is three times of the hole mobility throughout.

$N_A = 10^{16}/cm^3$ $\quad \rho_0 = 0.1\ \Omega cm$ $\quad \rho_1 = 0.03\ \Omega cm$ $\quad \mu_n = 3\mu_p$

$\sigma = \frac{1}{\rho} = q(n\mu_n + p\mu_p)$

$p = N_A = 10^{16}/cm^3 \Rightarrow \sigma_0 = pq\mu_p \quad (n \to small) = 10^{16}q\mu_p$

$\sigma_{new} = 10/3\,\sigma_{old}$

$n + N_A = p + N_D \Rightarrow n - p = N_D - N_A \quad np = n_i^2$

$nq(3\mu_p) + pq\mu_p = \frac{10^{17}}{3}q\mu_p \Rightarrow 9n + 3p = 10^{17} \quad np = 10^{20}$

$9n + 3\cdot\frac{10^{20}}{n} = 10^{17} \Rightarrow 9n^2 - 10^{17}n - 3\cdot10^{20} = 0 \Rightarrow n = \frac{10^{17} + \sqrt{10^{34} + 108\cdot10^{20}}}{18}$

$= \frac{10^{17}}{9} = 1.11\times10^{16}/cm^3$

$p = 9\times10^3 \quad n - p = N_D - N_A \Rightarrow 1.11\times10^{16} = N_D - 10^{16} \Rightarrow N_D = 2.11\times10^{16}$

5. (2 points) A particular semiconductor sample is doped with 2×10^{16} donor atoms/cm^3. The intrinsic carrier concentration (n_i) in the semiconductor is $2\times10^{14}/cm^3$ at 300 K.

Assuming n_i changes by one order of magnitude for every 100 K and complete ionization of the impurities, calculate the ratio of electron to hole concentration in this sample at 400 K.

$N_D = 2\times10^{16} = n$; $n_i\,(T = 400K) = 2\times10^{15} \Rightarrow p = 2\times10^{14} \Rightarrow \frac{n}{p} = 10^2$

Assuming n_i changes by one order of magnitude for every 100 K and complete ionization of the impurities, calculate the ratio of electron to hole concentration in this sample at 400 K.

6. (2 points) At room temperature, an n-type Si sample with doping concentration of 10^{17} cm^{-3} is being uniformly illuminated leading to uniform bulk generation of electron-hole-pairs (EHPs) with a rate of 10^{19} $cm^{-3}s^{-1}$. If the life time of hole is 1 μs, find out the separation between the quasi Fermi levels in eV.

7. (2 points) An n-type Si wafer with doping of 10^{17} cm^{-3} is uniformly illuminated with a generation rate of 10^{18} $cm^{-3}s^{-1}$. If the light source is switched off at t=0, calculate the time required (in μs) for the excess minority carriers to drop to 20 percent from its value at t=0. Assume complete ionization and τ_p=10^{-7}s.

8. (2 points) At 300 K, a Si sample is doped with 10^{16} cm^{-3} phosphorus atoms and 1.8 x 10^{16} cm^{-3} Boron atoms. The sample is uniformly illuminated leading to bulk generation of EHPs with a rate of 10^{18} cm^{-3} s^{-1}. Assume complete ionisation and τ_n=1 μs.

   (a) Find out the resistivity of the sample (in ohm-cm) before shining light on it.

   (b) Find out the reduction in resistivity of the illuminated sample (in μohm-cm) in steady state.

6. (2 points) At room temperature, an n-type Si sample with doping concentration of 10^{17} cm^{-3} is being uniformly illuminated leading to uniform bulk generation of electron-hole-pairs (EHPs) with a rate of 10^{19} cm^{-3}s^{-1}. If the life time of hole is 1 μs, find out the separation between the quasi Fermi levels in eV.

$N_D = 10^{17}$ $G_L = 10^{19}$ $\Delta p = \Delta n = 10^{19} \times 10^{-6} = 10^{13}$

$n = N_D + \Delta n = 10^{17}$ $p = p_0 + \Delta p = 10^3 + 10^{13} = 10^{13}$

$np = n_i^2 \exp\left(\frac{E_{fn} - E_{fp}}{KT}\right) \Rightarrow 10^{10} = \exp\left(\frac{E_{fn} - E_{fp}}{0.026}\right)$

$\Rightarrow 23.03 \times 0.026 = E_{fn} - E_{fp} = 0.599$ eV

7. (2 points) An n-type Si wafer with doping of 10^{17} cm^{-3} is uniformly illuminated with a generation rate of 10^{18} cm^{-3}s^{-1}. If the light source is switched off at t=0, calculate the time required (in μs) for the excess minority carriers to drop to 20 percent from its value at t=0. Assume complete ionization and $\tau_p = 10^{-7}$s.

$\Delta p(t=0) = 10^{17} \times 10^{-7} = 10^{10}$

$\frac{d(\Delta p)}{dt} = -\text{Recombination rate} = -\frac{\Delta p}{\tau_p} \Rightarrow \Delta p = \Delta p(t=0)\, e^{-t/\tau_p}$

$\Rightarrow 0.2 = e^{-t/\tau_p} \Rightarrow 1.61\tau_p = t = 1.61 \times 10^{-7} = 0.161\,\mu s$

8. (2 points) At 300 K, a Si sample is doped with 10^{16} cm^{-3} phosphorus atoms and 1.8 x 10^{16} cm^{-3} Boron atoms. The sample is uniformly illuminated leading to bulk generation of EHPs with a rate of 10^{18} cm^{-3} s^{-1}. Assume complete ionisation and τ_n=1 μs.

(a) Find out the resistivity of the sample (in ohm-cm) before shining light on it.

(b) Find out the reduction in resistivity of the illuminated sample (in μohm-cm) in steady state.

$N_D = 10^{16}$ $N_A = 1.8 \times 10^{16}$ $G_L = 10^{18}$ $\tau = 10^{-6}$

(a) $p_0 = N_A - N_D = 0.8 \times 10^{16}$ $n_0 = \frac{10^{20}}{0.8 \times 10^{16}} = 1.25 \times 10^4$

$\sigma_0 = q(p_0\mu_p + n_0\mu_n) = q p_0 \mu_p = (1.6 \times 10^{-19})(0.8 \times 10^{16})(450) = 0.576$ $(\Omega^{-1}cm^{-1})$

$\rho_0 = 1.736111 \; (\Omega^{-1}cm^{-1})$

(b) $\Delta n = 10^{18} \times 10^{-6} = 10^{12} = \Delta p \Rightarrow n = 10^{12}$ $p = 0.8 \times 10^{16}$

$\sigma_1 = 1.6 \times 10^{-19}\,(10^{12} \cdot 1350 + 10^{16} \cdot 0.8 \cdot 450) = 0.576216\ (\Omega^{-1}cm^{-1})$

$\Rightarrow \rho_1 = 1.7354\ (\Omega^{-1}cm^{-1})$

$\Rightarrow \Delta\rho = 700\mu\ (\Omega^{-1}cm^{-1})$

EE3001: Solid State Devices
TAs: Alias George (ee24d402@smail.iitm.ac.in) and Devendar K (ee24s015@smail.iitm.ac.in)
Assignment #4

Try yourself to solve the problems with detailed steps. Take a limited time quiz to test your understanding on related concepts.

Problems will be discussed in class. You need to contact TAs or teacher if you are unable to do anything useful by yourself or with the help of your friends.

If needed, assume the following values by default unless otherwise specified: Si is the default semiconductor; SiO_2 is the default oxide; T @room temperature is 300 K; n_i@room temp. = $10^{10} cm^{-3}$; k_BT=0.026 eV (at room temperature); $E_{G,Si}$ = 1.12 eV and temperature independent; at 300 K, N_c=$2.8 \times 10^{19} cm^{-3}$ and N_v=$1.02 \times 10^{19} cm^{-3}$

1. (5 points) Light is shone on the surface of an n-type silicon wafer generating 10^{16} electron-hole pairs per cm^2 per second only on this surface. Assuming T = 300K, μ_n= 1350 cm^2/V-sec, μ_p= 450 cm^2/V-sec, and τ_p=τ_n=1 μs what is the electron current density (in mA/cm^2), at a distance of 5 μm from this surface?

2. (5 points) Fig.1 shows a non-uniformly doped n-type semiconductor block with 3 distinct regions. For $x < 0$, $N_D = 10^{14}$ $/cm^3$ and for $x > l$, $N_D = 10^{18}$ $/cm^3$. For $0 < x < l$, the donor concentration increases exponentially and can be expressed as

   $$N_D(x) = N_{D0} e^{ax},$$

   where N_{D0} and a are constants. Assuming $l = 1$ μm, plot the electric field variation in the semiconductor block at room temperature (300 K).

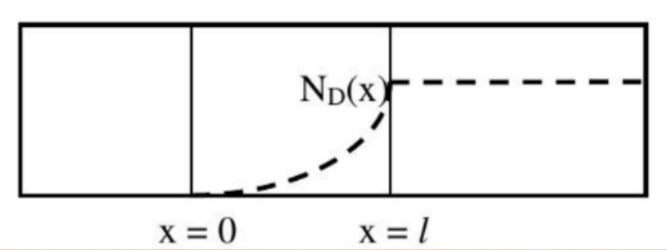


Figure 1: A bar of n-type semiconductor showing graded impurity distribution.

3. (5 points) Fig. 2 shows a sample used for the Hall Effect experiment, where $W = 0.1$ mm, $d = 1$ mm and $L = 1$ cm. A current of $I = 1$ A flows through the sample when $V_{CD} = 50$ V. If the measured Hall voltage is $V_H = -6.25$ mV when the current flowing through the sample is $I = 1$ A and the applied magnetic field strength is $B_z = 10^{-5}$ Wb/cm^2 , what is

2. (5 points) Fig.1 shows a non-uniformly doped n-type semiconductor block with 3 distinct regions. For $x < 0$, $N_D = 10^{14}\ /cm^3$ and for $x > l$, $N_D = 10^{18}\ /cm^3$. For $0 < x < l$, the donor concentration increases exponentially and can be expressed as

$$N_D(x) = N_{D0}e^{ax},$$

where N_{D0} and a are constants. Assuming $l = 1\ \mu$m, plot the electric field variation in the semiconductor block at room temperature (300 K).

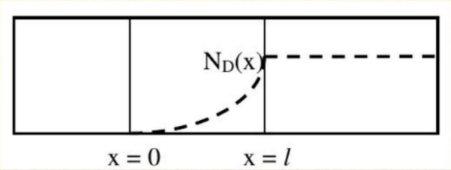


Figure 1: A bar of n-type semiconductor showing graded impurity distribution.

$N_{D0} = 10^{14}$

$N_{D0} e^{al} = 10^{18}$

$\Rightarrow e^{al} = 10^4$

$\Rightarrow al = 9.21$

$\Rightarrow a = 9.21 \times 10^6\ m^{-1}$

$V_T = \frac{KT}{q} = 0.026V$

$J_n = J_p = 0$

$J_n(x) = q\mu_n n(x) \mathcal{E}(x) + qD_n \frac{dn(x)}{dx} = 0$

$\Rightarrow \mathcal{E}(x) = -\frac{D_n}{\mu_n} \cdot \frac{1}{n(x)} \frac{dn(x)}{dx}$

$D_n = \mu_n V_T \Rightarrow \mathcal{E}(x) = -\frac{V_T}{n(x)} \cdot \frac{dn(x)}{dx}$

$x < 0 : \mathcal{E}(x) = 0 \qquad x > l : \mathcal{E}(x) = 0$

$0 < x < l : \mathcal{E}(x) = -\frac{V_T}{n(x)} \quad an(x) = aV_T \Rightarrow$ constant

$= -2394.69\ V/cm$

1. (5 points) Light is shone on the surface of an n-type silicon wafer generating 10^{16} electron-hole pairs per cm^2 per second only on this surface. Assuming T = 300K, μ_n= 1350 cm^2/V-sec, μ_p= 450 cm^2/V-sec, and τ_p=τ_n=1 μs what is the electron current density (in mA/cm^2), at a distance of 5 μm from this surface?

$D_p = \mu_p \cdot \frac{KT}{q} = 450\ cm^2/Vsec \cdot 0.026 = 11.7\ cm^2/s$

$L_p^2 = 11.7 \times 10^{-6} \Rightarrow 3.42 \times 10^{-3}\ cm = L_p$

$\Delta p_x = \Delta p_0 e^{-x/L_p} \qquad 10^{16} \times 10^{-6} = \Delta n(0)$

$J_p(x) = qD_p \frac{d(\Delta p_x)}{dx} = qD_p \frac{e^{-x/L_p}}{-L_p} \Delta n_0 = \frac{1.6 \times 10^{-19} \times 11.7 \times 10^{10} \cdot 0.863}{-3.42 \times 10^{-3}}$

$= -4.72 \times 10^{-6}$

$\Rightarrow J_n(x) = 4.72 \times 10^{-6}\ A/cm$

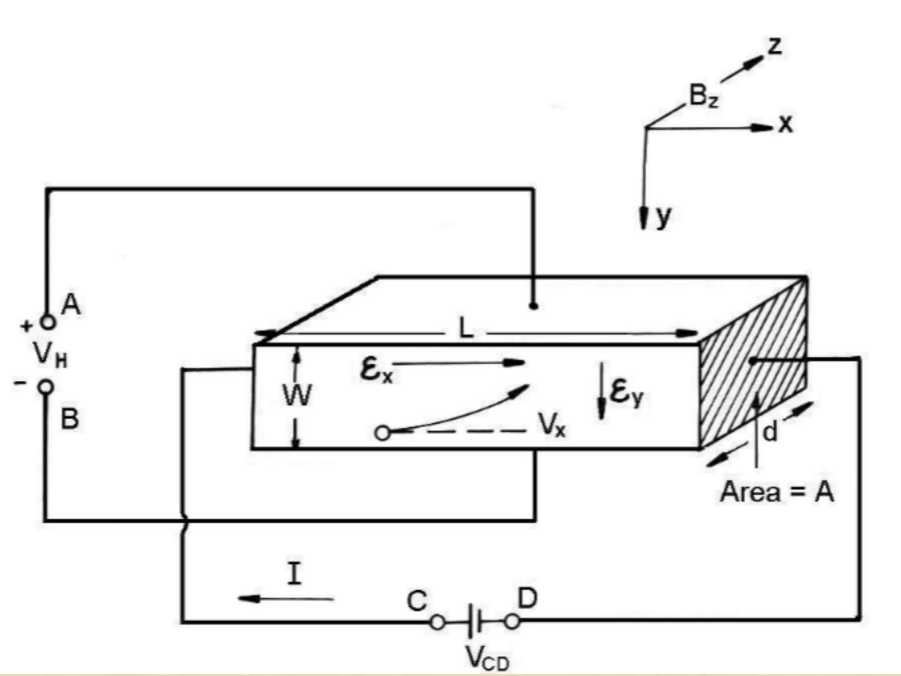


Figure 2: Hall experiment setup.

(a) the type of the semiconductor,

(b) the doping concentration (assuming all dopants are ionized) and

(c) the mobility of the majority carriers in the sample?

(a) n-type (-ve hall coeff)

(b) $V_H/\omega = E_y = V_x B_z$ (Florentz = 0) $\quad J = nqV_x \quad V_x = \frac{J}{nq} = \frac{I}{Anq}$

$$\frac{V_H}{\omega} = \frac{IB_z}{Anq} \Rightarrow n = \frac{IB_z\omega}{V_H A q} = \frac{1\times10^{-5}\times10^{-2}}{6.25\times10^{-3}\times10^{-3}\times1.6\times10^{-19}} = 10^{17}/cm^3$$

(c) $\sigma = qn\mu_n = J/E = \frac{I/A}{V/L} = \frac{1000}{50} = 20\ (\Omega cm)^{-1}$

$$\mu_n = \frac{\sigma}{nq} = \frac{20}{1.6\times10^{-2}} = 1250\ cm^2/V.sec$$

$\epsilon_{Si} = 1.0443 \times 10^{-12}$

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TAs: Alias George (ee24d402@smail.iitm.ac.in) and Devendar K (ee24s015@smail.iitm.ac.in)
Assignment #5

Try yourself to solve the problems with detailed steps. Take a limited time quiz to test your understanding on related concepts.

Problems will be discussed in class. You need to contact TAs or teacher if you are unable to do anything useful by yourself or with the help of your friends.

If needed, assume the following values by default unless otherwise specified: Si is the default semiconductor; SiO_2 is the default oxide; T @room temperature is 300 K; n_i@room temp. = $10^{10} cm^{-3}$; k_BT=0.026 eV (at room temperature); $E_{G,Si}$ = 1.12 eV and temperature independent; at 300 K, N_c=$3.2 \times 10^{19} cm^{-3}$, N_v=$1.8 \times 10^{19} cm^{-3}$, μ_n = $1350 cm^2/V.s$, μ_p = $450 cm^2/V.s$; Permittivity for vacuum: ϵ_0=8.85×10^{-14}F/cm, relative permittivity: $\epsilon_{r,Si}$=11.8, ϵ_{r,SiO_2}=3.9; pn diode dopants are completely ionized.

1. (2 points) Acceptor and donor concentration in the p- and n-side of an abrupt pn-junction are $N_A = 4 \times 10^{18}\ /cm^3$ and $N_D = 4 \times 10^{17}\ /cm^3$, respectively.

   (a) Sketch (in scale) the electric field distribution from $-X_{p0}$ to $+X_{n0}$ considering the metallurgical junction at the origin. Here X_{p0} and X_{n0} are the penetration depth of the space charge region in p and n regions, respectively.

   (b) Now if p-side doping is modified to $N_A = 8 \times 10^{18}\ /cm^3$, sketch the electric field profile on the same plot.

   (c) How will the plot in (a) appear at an elevated temperature?

2. (2 points) The built-in potential of an abrupt p-n junction is 0.75 V. If the depletion layer width (W) and junction capacitance (C_J) are 1 μm and 10 pF when the reverse bias (V_r) is 1.25 V, what are the values of (a) C_J and (b) W when V_r is 31.25 V?

3. (2 points) At T=300 K, measured voltage dependent inverse squared junction capacitance ($1/C_j^2$) of an abrupt Si p^{++}n junction with cross sectional area of 100 μm^2 predicts an x-intercept of 0.725 V and y-intercept of $10^{30}\ F^{-2}$. If the n-sided doping is given by $M \times 10^{14}\ cm^{-3}$, find the value of M.

4. (2 points) A pn diode has N_A=$10^{16}\ cm^{-3}$ and N_D=$10^{15}\ cm^{-3}$. Another identically doped pn diode has an additional intrinsic layer introduced in between the p- and n-sides leading to a p-i-n structure. The intrinsic or i-layer in the p-i-n structure has a length of 2 μm.

   (a) Under zero bias, calculate the ratio of the depletion capacitance of the p-i-n ($C_{j,pin}(0)$) and p-n ($C_{j,pn}(0)$) diodes.

   (b) Under -5 V reverse bias, calculate the depletion capacitances of the p-i-n ($C_{j,pin}(-5)$) and p-n ($C_{j,pn}(-5)$) diodes.

1. (2 points) Acceptor and donor concentration in the p- and n-side of an abrupt pn-junction are $N_A = 4 \times 10^{18}\ /cm^3$ and $N_D = 4 \times 10^{17}\ /cm^3$, respectively.

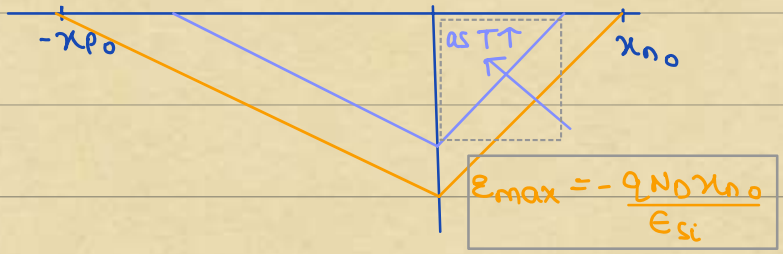
   (a) Sketch (in scale) the electric field distribution from $-X_{p0}$ to $+X_{n0}$ considering the metallurgical junction at the origin. Here X_{p0} and X_{n0} are the penetration depth of the space charge region in p and n regions, respectively.

   (b) Now if p-side doping is modified to $N_A = 8 \times 10^{18}\ /cm^3$, sketch the electric field profile on the same plot.

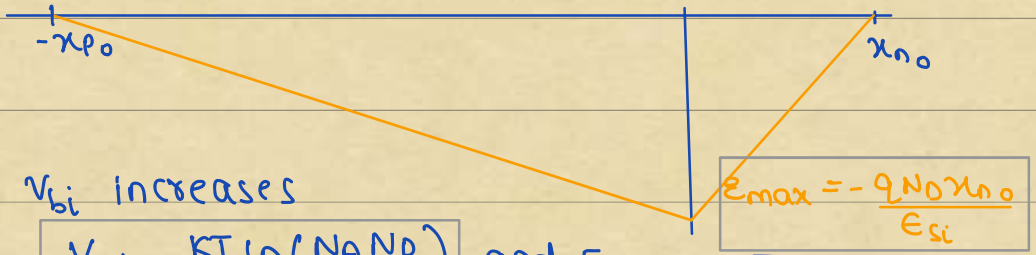
   (c) How will the plot in (a) appear at an elevated temperature?

$q N_A x_{p_0} = q N_D x_{n_0} \Rightarrow x_{p_0} = 10\, x_{n_0}$

a)



b) $x_{p_0} = 20\, x_{n_0}$



V_{bi} increases

$V_{bi} = \frac{KT}{q} \ln\left(\frac{N_A N_D}{n_i^2}\right)$ and $E_{max} \approx \sqrt{V_{bi}}$

c) as T↑ ; $\frac{KT}{q}$↑ but n_i↑ ; so V_{bi} decreases,

and since $W \propto \sqrt{V_{bi}}$ so W decreases

and also E_{max} decreases

2. (2 points) The built-in potential of an abrupt p-n junction is 0.75 V. If the depletion layer width (W) and junction capacitance (C_J) are 1 μm and 10 pF when the reverse bias (V_r) is 1.25 V, what are the values of (a) C_J and (b) W when V_r is 31.25 V?

$V_{bi} = 0.75 \quad V_r = 1.25 \quad W = 1\,\mu m \quad C_j = 10\,pF$

$W \propto \sqrt{V_{bi} + V_r} \Rightarrow \frac{W_2}{W_1} = \frac{\sqrt{32}}{\sqrt{2}} = 4$

$\Rightarrow W_2 = 4\mu m$

$C_j \propto \frac{1}{W} \Rightarrow \frac{C_2}{C_1} = \frac{W_1}{W_2} \Rightarrow C_2 = 2.5\,pF$

3. (2 points) At T=300 K, measured voltage dependent inverse squared junction capacitance ($1/C_j^2$) of an abrupt Si p^{++}n junction with cross sectional area of 100 μm^2 predicts an x-intercept of 0.725 V and y-intercept of $10^{30}\ F^{-2}$. If the n-sided doping is given by $M \times 10^{14}\ cm^{-3}$, find the value of M.

$C_j = \frac{\epsilon_{Si} A}{W}$ $\qquad W = \sqrt{\frac{2\epsilon_{Si}(V_{bi}-V)}{qN_{eff}}}$

$\Rightarrow \frac{1}{C_j^2} = \left(\frac{2\epsilon_{Si}}{qN_{eff}\epsilon_{Si}^2 A^2}\right)(V_{bi}-V)$

$\frac{1}{C_j^2} = \frac{2V_{bi}}{qN_{eff}\epsilon_{Si}A^2} - \left(\frac{2}{qN_{eff}\epsilon_{Si}A^2}\right)V$

$V_{bi} = 0.725\,V \qquad 10^{30} = \frac{2V_{bi}}{qN_{eff}\epsilon_{Si}A^2}$

$\Rightarrow 10^{30} = \frac{2 \times 0.725}{1.6\times10^{-19} \times N_{eff} \times 10^{-12} \times 1.0443\times10^{-12}}$

$\Rightarrow N_{eff} = 8.678\times10^{12}$

$p^{++}n$

$N_A >>> N_D \Rightarrow \frac{1}{N_A} \approx 0$

$\frac{1}{N_{eff}} = \frac{1}{N_A} + \frac{1}{N_D}$

$\Rightarrow N_D = N_{eff} = 0.0868 \times 10^{14}$

4. (2 points) A pn diode has N_A=10^{16} cm^{-3} and N_D=10^{15} cm^{-3}. Another identically doped pn diode has an additional intrinsic layer introduced in between the p- and n-sides leading to a p-i-n structure. The intrinsic or i-layer in the p-i-n structure has a length of 2 μm.

(a) Under zero bias, calculate the ratio of the depletion capacitance of the p-i-n ($C_{j,pin}(0)$) and p-n ($C_{j,pn}(0)$) diodes.

(b) Under -5 V reverse bias, calculate the depletion capacitances of the p-i-n ($C_{j,pin}(-5)$) and p-n ($C_{j,pn}(-5)$) diodes.

(c) Find the ratios ($C_{j,pn}(-5)/C_{j,pn}(0)$) and ($C_{j,pin}(-5)/C_{j,pin}(0)$).

(d) Name an application where such p-i-n structure can be useful.

$V_{bi} = \frac{KT}{q} \ln\left(\frac{N_A N_D}{n_i^2}\right) = 0.026 \ln\left(\frac{10^{31}}{10^{20}}\right) = 0.656\,V$

$V_R = 0 \Rightarrow W_{pn}(0) = \sqrt{\frac{2(1.04\times10^{-12})(0.656)}{1.6\times10^{-19}\times 9.1\times10^{14}}} = 0.95\,\mu m$

$W_{pin}(0) = \sqrt{2^2 + 0.95^2} = 2.21\,\mu m$

a) $\frac{C_{j,pin}(0)}{C_{j,pn}(0)} = \frac{W_{pn}(0)}{W_{pin}(0)} = \frac{0.95}{2.21} = 0.429$

b) $W_{pn}(-5) = W_{pn}(0)\sqrt{\frac{V_{bi}+5}{V_{bi}}} = 0.95\sqrt{\frac{5.656}{0.656}} = 2.79\,\mu m$

$C_{j,pn}(-5) = \frac{\epsilon_{Si}}{W_{pn}(-5)} = 3.73\,nF/cm^2$

$W_{pin}(-5) = 3.43\,\mu m \Rightarrow C_{j,pin}(-5) = \frac{\epsilon_{Si}}{W_{pin}(-5)} = 3.03\,nF/cm^2$

(c) $\frac{C_{j,pn}(-5)}{C_{j,pn}(0)} = 0.34$ 66% drop $\quad \frac{C_{j,pin}(-5)}{C_{j,pin}(0)} = 0.61$ 39% drop $\Rightarrow$ more stable

(d) Photodetectors: i-layer provides large intrinsic volume where incoming photons can be absorbed to create electron hole pairs

(c) Find the ratios $(C_{j,pn}(-5)/C_{j,pn}(0))$ and $(C_{j,pin}(-5)/C_{j,pin}(0))$.

(d) Name an application where such p-i-n structure can be useful.

5. (2 points) A p^+n diode has non-uniform n-sided doping given by $N_D(x)$=$N_{D0}(x/x_0)^m$ cm^{-3}. Such a diode is being used as a bias-dependent variable capacitor $(C(V))$ in a local oscillator circuit producing a variable resonant frequency $f_r = 1/2\pi\sqrt{LC(V)}$. It is found that $f_r \propto V^n$.

   (a) Should the diode be used in forward bias or reverse bias?

   (b) If $m = 0$, find the value of n.

   (c) If $m = 1$, find the value of n.

   (d) What should be the value of m in order to ensure $n = 1$?

5. (2 points) A p^+n diode has non-uniform n-sided doping given by $N_D(x)=N_{D0}(x/x_0)^m$ cm^{-3}. Such a diode is being used as a bias-dependent variable capacitor ($C(V)$) in a local oscillator circuit producing a variable resonant frequency $f_r = 1/2\pi\sqrt{LC(V)}$. It is found that $f_r \propto V^n$.

(a) Should the diode be used in forward bias or reverse bias?

(b) If $m = 0$, find the value of n.

(c) If $m = 1$, find the value of n.

(d) What should be the value of m in order to ensure $n = 1$?

a) $C_j \rightarrow$ junction capacitance $C_d \rightarrow$ diffusion capacitance

Forward bias: C_d is dominant $C_d \propto I_f \propto e^{qV/kT}$

small voltage fluctuations in voltage lead to massive capacitance swings.

Reverse bias: C_j is dominant. $C_j(V) = \frac{\epsilon_{Si}A}{W} \propto (V_{bi}+V_R)^{\frac{-1}{m+2}}$

stable, near infinite resistance $\Rightarrow$ oscillator loses very little energy to heat.

$$\frac{d^2\phi}{dx^2} = -\frac{dE}{dx} = -\frac{\rho(x)}{\epsilon_{Si}} = -\frac{qN_D(x)}{\epsilon_{Si}} \Rightarrow \frac{dE}{dx} = \frac{qN_{D0}x^m}{\epsilon_{Si}x_0^m}$$

$$\Rightarrow E(x) = \frac{-qN_{D0}}{\epsilon_{Si}x_0^m(m+1)}(W^{m+1}-x^{m+1})$$

$$V = -\int_0^W E(x)\,dx = \frac{qN_{D0}}{\epsilon_{Si}x_0^m}\frac{W^{m+2}}{(m+1)(m+2)} \Rightarrow W \propto V^{1/m+2}$$

$$\Rightarrow C \propto V^{-1/m+2}$$

$$f \propto V^{1/2(m+2)} \Rightarrow n = \frac{1}{2(m+2)}$$

b) $n = 1/4$ c) $n = 1/6$ d) $m = -1.5$